

Technology Stack

Date	26 June 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, Gradio	Delivers a clean, mobile-friendly interface where applicants can enter their details and view prediction outcomes.
2	Backend / Server-Side	Python, Gradio, Scikit-learn	Manages core business logic, handles incoming requests, runs the trained ML model, and returns live approval predictions.
3	Database / Data Storage	SQLite / CSV Dataset	Efficiently stores applicant records and historical data using a lightweight storage solution.
4	Cloud / Hosting / Deployment	IBM Watson Machine Learning, Gradio	Hosts the trained model in the cloud, enabling secure, real-time predictions through the web app.
5	Version Control & CI/CD	Git, GitHub	Tracks code changes, supports collaboration among team members, and streamlines ongoing maintenance.
6	Third-Party APIs / Other Tools	Pandas, NumPy, Matplotlib, Joblib	Enables data cleaning, numerical operations, visualizations, and saving/loading the trained model.